MA2031- Linear Algebra for Engineers

Sumesh K

Module-5 (Lecture note-5)

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Positive semidefinite matrices

Definition: Let $A \in M_n(\mathbb{C})$. Then A is said to be positive semidefinite if

$$\langle Ax, x \rangle \geq 0$$

for all $x \in \mathbb{C}^n$.

- Let $A = \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$. Then for all $x = (x_1, x_2)^t \in \mathbb{C}^2$

$$\langle Ax, x \rangle = (x_1 - x_2)\overline{(x_1 - x_2)} \geq 0.$$

  Hence, A is a positive semidefinite matrix.

- Let $A = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}$. Then for $x = (1, -3)^t \in \mathbb{C}^2$

$$\langle Ax, x \rangle = -2 < 0.$$

  Hence, A is **not** a positive semidefinite matrix.

Theorem: Let $A \in M_n(\mathbb{C})$. Then the following conditions are equivalent:

1. A is positive semidefinite.
2. $A = A^*$ and $\sigma(A) \subseteq [0, \infty)$.
3. $A = C^2$ for some $C = C^*$.
4. $A = B^*B$ for some $B \in M_n(\mathbb{C})$.
5. There exists $\{v_1, \cdots, v_n\} \subseteq \mathbb{C}^n$ such that $A = \big[\langle v_j, v_i \rangle\big]$.

Proof: $(1) \Rightarrow (2)$ Given that $\langle Ax, x\rangle \geq 0$ for all $x \in \mathbb{C}^n$. Then for all $x, y \in \mathbb{C}^n$,

$$\begin{aligned}\langle Ax, y\rangle &= \frac{1}{4}\sum_{k=0}^{3}(i)^k\big\langle A(x+i^k y), x+i^k y\big\rangle = \frac{1}{4}\sum_{k=0}^{3}(i)^k\overline{\big\langle A(x+i^k y), x+i^k y\big\rangle} \\ &= \frac{1}{4}\sum_{k=0}^{3}(i)^k\big\langle x+i^k y, A(x+i^k y)\big\rangle = \langle x, Ay\rangle.\end{aligned}$$

Hence $A = A^*$. Now if $\lambda \in \sigma(A)$ and $0 \neq x \in \mathbb{C}^n$ such that $Ax = \lambda x$. Then

$$\lambda\langle x, x\rangle = \langle \lambda x, x\rangle = \langle Ax, x\rangle \geq 0,$$

hence $\lambda \geq 0$.

Proof continues..

$(2) \Rightarrow (3)$ Since $A = A^*$, by spectral theorem, $A = UDU^*$ for some $U \in M_n(\mathbb{C})$ unitary and $D = diag\{d_1, d_2, \cdots, d_n\} \in M_n(\mathbb{C})$ with $d_i \in \sigma(A) \subseteq [0, \infty)$. Let $D' := diag\{\sqrt{d_1}, \sqrt{d_2}, \cdots, \sqrt{d_n}\} \in M_n(\mathbb{C})$ and $C := UD'U^*$. Then

$$C^2 = (UD'U^*)(UD'U^*) = U(D')^2U^* = UDU^* = A.$$

$(3) \Rightarrow (4)$ Take $B = C$.
$(4) \Rightarrow (5)$ Suppose $B = [v_1|v_2|\cdots|v_n]$ where $v_i \in \mathbb{C}^n$. Observe that $A = B^*B = \big[\langle v_j, v_i\rangle\big]$.
$(5) \Rightarrow (1)$ Given that $A = \big[\langle v_j, v_i\rangle\big]$, where $v_i \in \mathbb{C}^n$. Then for all $x = [x_1, \cdots, x_n]^t \in \mathbb{C}^n$,

$$\langle Ax, x\rangle = \left\langle \begin{bmatrix} \langle v_1, v_1\rangle & \cdots & \langle v_n, v_1\rangle \\ \vdots & & \vdots \\ \langle v_1, v_n\rangle & \cdots & \langle v_n|v_n\rangle \end{bmatrix} \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}, \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} \right\rangle = \left\langle \begin{bmatrix} \sum_{i=1}^n \langle v_i, v_1\rangle x_i \\ \vdots \\ \sum_{i=1}^n \langle v_i, v_n\rangle x_i \end{bmatrix}, \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} \right\rangle$$

$$= \sum_{i,j=1}^n \langle v_i, v_j\rangle x_i \overline{x_j} = \sum_{i,j=1}^n \big\langle x_i v_i, x_j v_j \big\rangle$$

$$= \Big\langle \sum_{i=1}^n x_i v_i, \sum_{j=1}^n x_j v_j \Big\rangle \geq 0.$$

This completes the proof.

Exercises

1. Determine which of the following matrices are positive semidefinite.

   (a) $A = \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$

   (b) $A = \begin{bmatrix} 1 & 2 \\ 2 & 0 \end{bmatrix}$

   (c) $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 1 & 0 \\ 3 & 0 & 0 \end{bmatrix}$

   (c) $A = \begin{bmatrix} 3 & 1 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$

2. Suppose $A \in M_n(\mathbb{C})$ is a positive semidefinite matrix. Show that there exists a positive semidefinite matrix B such that $A = B^2$. (In fact, this matrix B is unique and is called the square root A.)

   (Hint: use spectral theorem.)